

1 Solution — GIAM 6.1

Problem 1

1.1 Problem

The *lexicographic order*, $<_{\text{lex}}$, is a relation on the set of all words, where $x <_{\text{lex}} y$ means that x would come before y in the dictionary. Consider just the three-letter words like “iff”, “fig”, “the”, et cetera. Come up with a usable definition for $x_1x_2x_3 <_{\text{lex}} y_1y_2y_3$.

1.2 Setup

Assume the 26 letters of the alphabet carry their familiar total order

$$\mathbf{a} < \mathbf{b} < \mathbf{c} < \cdots < \mathbf{z},$$

denoted simply by $<$. A *three-letter word* is an ordered triple $x_1x_2x_3$ with each $x_i \in \{\mathbf{a}, \mathbf{b}, \dots, \mathbf{z}\}$.

1.3 Definition

For three-letter words $x_1x_2x_3$ and $y_1y_2y_3$, define

$$\begin{aligned} x_1x_2x_3 <_{\text{lex}} y_1y_2y_3 \iff & x_1 < y_1, \quad \text{or} \\ & x_1 = y_1 \text{ and } x_2 < y_2, \quad \text{or} \\ & x_1 = y_1, x_2 = y_2 \text{ and } x_3 < y_3. \end{aligned}$$

Equivalently, in compact form,

$$x_1x_2x_3 <_{\text{lex}} y_1y_2y_3 \iff \exists k \in \{1, 2, 3\} : x_k < y_k \text{ and } x_j = y_j \text{ for all } j < k.$$

In words: scan the two words from left to right; the *first position at which they differ* decides the order, with the word holding the alphabetically earlier letter at that position coming first. If no such position exists, the two words are equal.

1.4 Why the Definition Is Usable

Given any two three-letter words, the definition is a finite, mechanical check:

1. Compare x_1 with y_1 . If they differ, the word with the smaller first letter is smaller.
2. Otherwise, compare x_2 with y_2 . If they differ, the word with the smaller second letter is smaller.
3. Otherwise, compare x_3 with y_3 . If they differ, the word with the smaller third letter is smaller.
4. If all three positions agree, the two words are equal, so neither is $<_{\text{lex}}$ the other.

The relation inherits the basic properties of dictionary order:

- **Irreflexivity.** $x_1x_2x_3 \not<_{\text{lex}} x_1x_2x_3$, since no $x_i < x_i$.
- **Trichotomy.** For any pair of words, exactly one of $x <_{\text{lex}} y$, $x = y$, or $y <_{\text{lex}} x$ holds — singled out by the first index at which they disagree (or by the absence of such an index).
- **Transitivity.** If $x <_{\text{lex}} y$ and $y <_{\text{lex}} z$, then $x <_{\text{lex}} z$: let i be the first index where x and y differ, and j the first where y and z differ. The first index where x and z differ is at most $\min(i, j)$, and the inequality at that position carries over.

Hence $<_{\text{lex}}$ is a *strict total order* on the set of three-letter words.

1.5 Examples

Each comparison is decided at the first index where the two words differ.

- **fig** $<_{\text{lex}}$ **iff**: differ at index 1, and **f** $<$ **i**.
- **iff** $<_{\text{lex}}$ **the**: differ at index 1, and **i** $<$ **t**.
- **cat** $<_{\text{lex}}$ **cow**: agree at index 1; differ at index 2, and **a** $<$ **o**.
- **run** $<_{\text{lex}}$ **rut**: agree at indices 1 and 2; differ at index 3, and **n** $<$ **t**.